

Date Fruit Variety & Quality Models: Evaluation Report

Generated 2026-09-29 from the metric files in `outputs/` · models: effv2b0_260, grade_mnv3+grade_effv2b0, maturity_effv2b0

Summary of results (held-out test sets)

Component	Model	Test n	Accuracy	Macro-F1	Loss	Accuracy with manual review of low-confidence
Phase 1 · Variety classification	EfficientNetV2-B0 260	275	97.5%	0.972	0.196	99.2% on 95% auto-accepted
Phase 2 · Quality grade (1–3)	Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260	451	84.5%	0.813	0.371	97.6% on 47% auto-accepted
Phase 2 · Size class	Random forest on outline geometry	330	80.3%	–	–	–
Phase 2 · Maturity stage	EfficientNetV2-B0 260	463	67.8%	0.667	0.881	81.2% on 66% auto-accepted

Model choice, calibration temperature and review thresholds were all fixed on validation splits; each test set was used once for the numbers shown. Loss is categorical cross-entropy of the uncalibrated model with test-time augmentation. Colour uniformity, defects, skin and weight are measurement rules without labelled data (section 6).

Phase 1 · Variety classification

Random stratified 70/15/15 split of 10 varieties (Zenodo web photos + Alhamdan studio photos).

97.5%
test accuracy

0.972
macro-F1

0.974
weighted-F1

0.196
test loss (cross-entropy)

99.3%
top-2 accuracy

275
test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	90.2%	0.918	0.269	91.3%
EfficientNetV2-B0 260 ✓ selected	97.5%	0.978	0.186	97.5%

Selected on validation: **EfficientNetV2-B0 260**. ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

Per-class results (test)

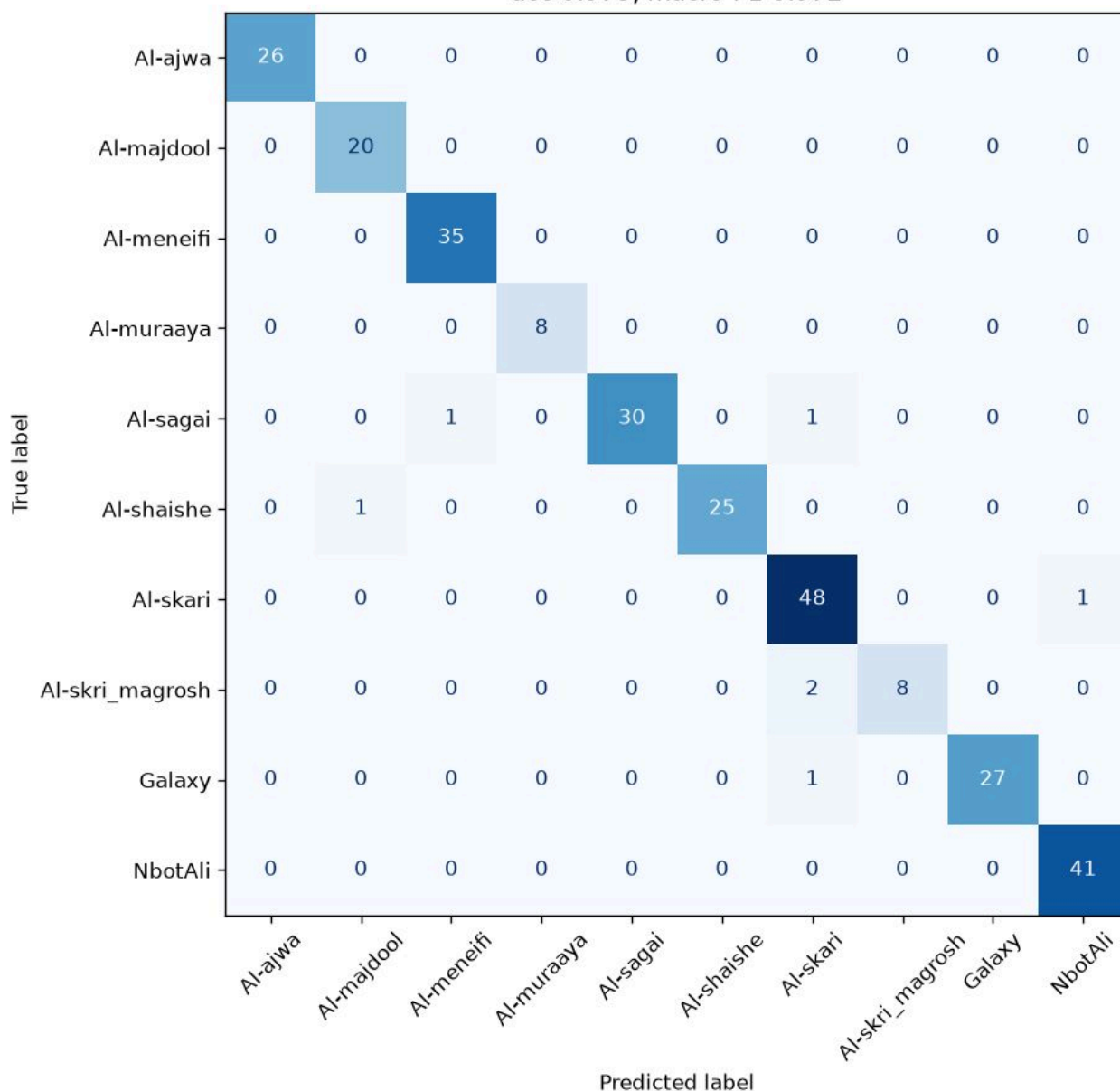
Class	Precision	Recall	F1	n
Al-ajwa	1.000	1.000	1.000	26
Al-majdool	0.952	1.000	0.976	20
Al-meneifi	0.972	1.000	0.986	35
Al-muraaya	1.000	1.000	1.000	8
Al-sagai	1.000	0.938	0.968	32
Al-shaishe	1.000	0.962	0.980	26
Al-skari	0.923	0.980	0.950	49
Al-skri_magrosh	1.000	0.800	0.889	10
Galaxy	1.000	0.964	0.982	28
NbotAli	0.976	1.000	0.988	41

Accuracy by source

Source / variety	Accuracy
alhamdan	98.2%
zenodo	93.9%

Confusion matrix (test)

test confusion matrix (effv2b0_260_tta)
acc 0.975, macro-F1 0.972



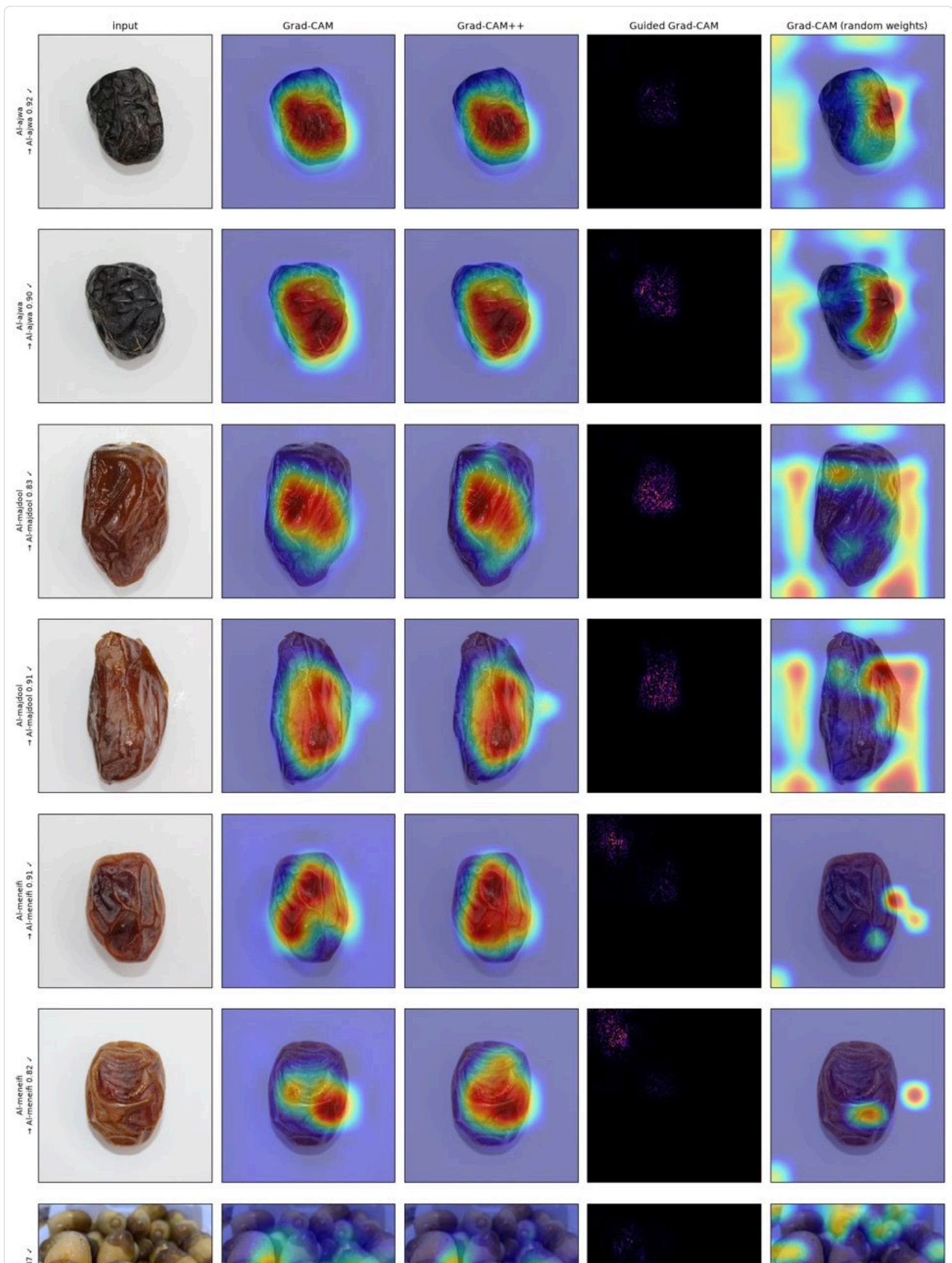
Calibration and manual-review threshold

Temperature scaling ($T = 0.5162$) fitted on validation. Expected calibration error on test $0.121 \rightarrow \mathbf{0.013}$; test NLL $0.196 \rightarrow 0.082$. The review threshold (0.787) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **94.9%** of images at **99.2%** accuracy and sends 14 of 275 to manual review.

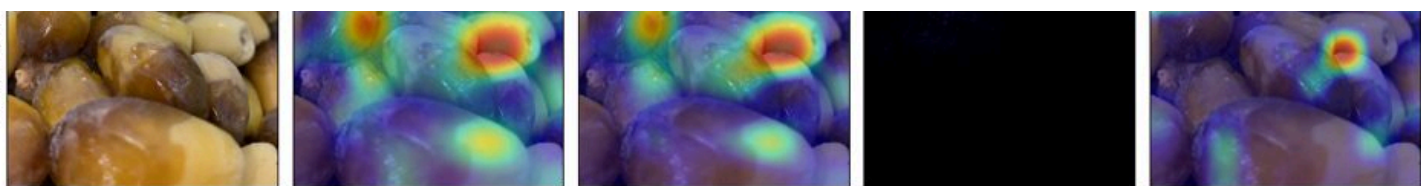
Confidence $\geq$	Auto-accepted	Accuracy on accepted
0.00	100.0%	97.5%
0.50	98.5%	98.2%
0.60	97.8%	98.5%
0.70	96.4%	98.5%
0.80	94.9%	99.2%
0.85	93.5%	99.6%
0.90	92.7%	99.6%
0.95	89.5%	99.6%
0.98	85.5%	99.6%

Explainability: Grad-CAM sanity checks (test split)

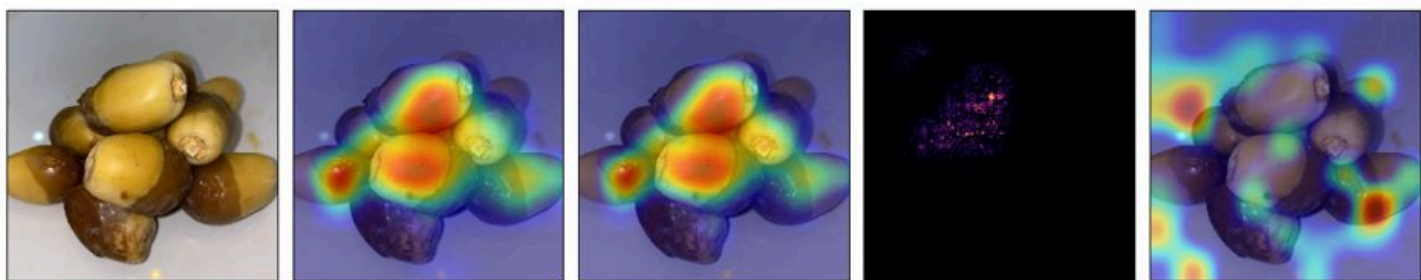
Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	0.013	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.428 vs 0.244	pass
Share of heatmap energy on fruit pixels vs fruit share of image	74.7% vs 28.3%	pass



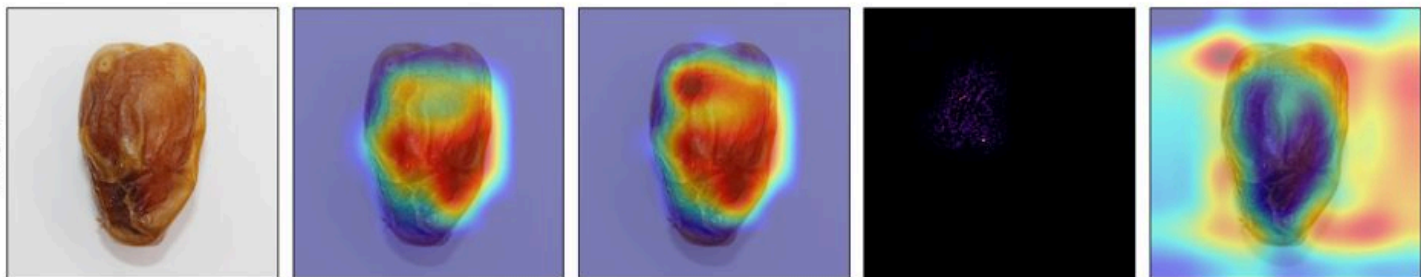
Al-muraaya
→ Al-muraaya 0.2



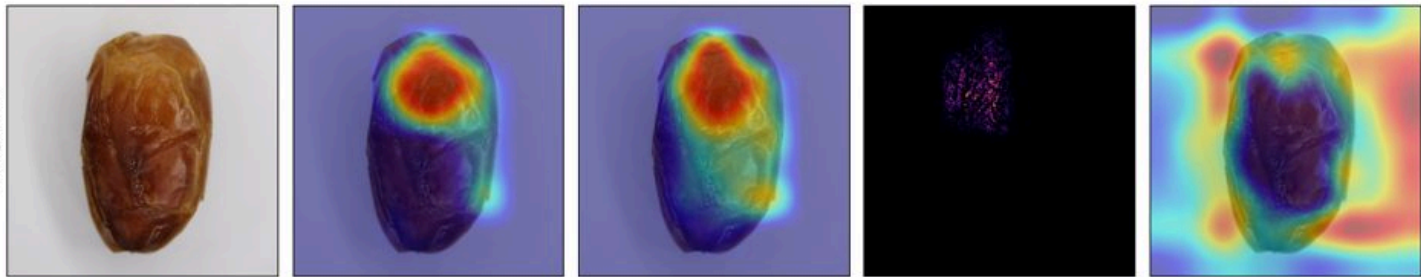
Al-muraaya
→ Al-muraaya 0.92 ✓



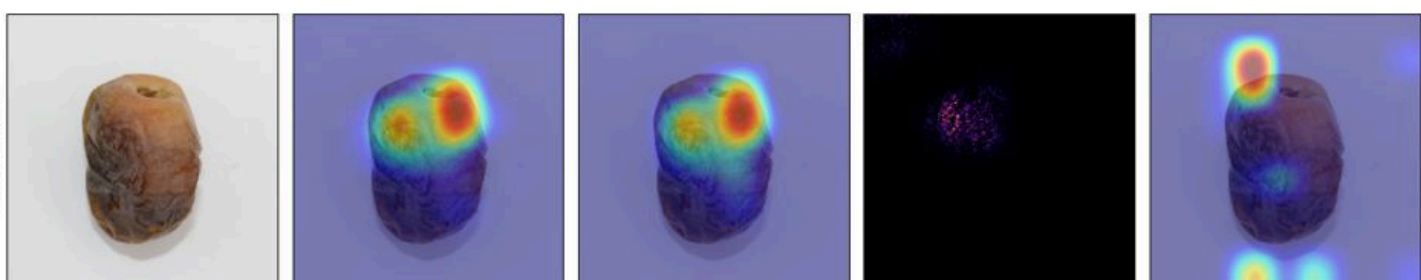
Al-sagali
→ Al-sagali 0.92 ✓



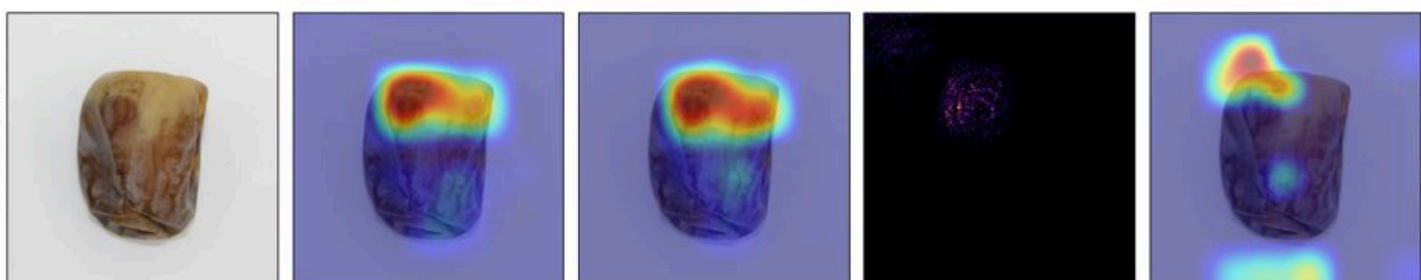
Al-sagali
→ Al-sagali 0.54 ✓



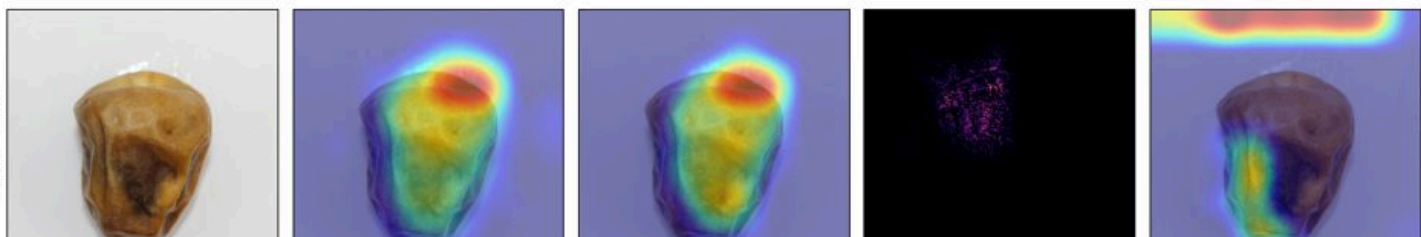
Al-shaiche
→ Al-shaiche 0.82 ✓



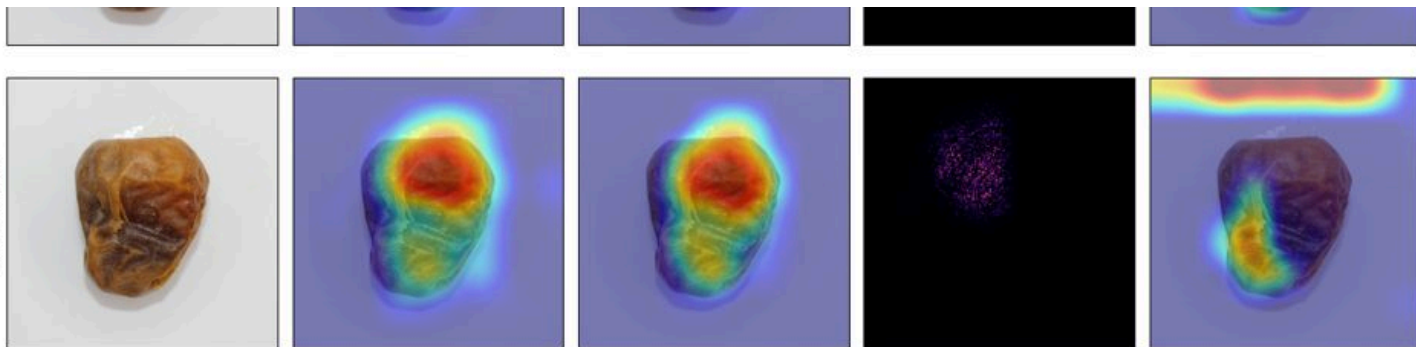
Al-shaiche
→ Al-shaiche 0.83 ✓



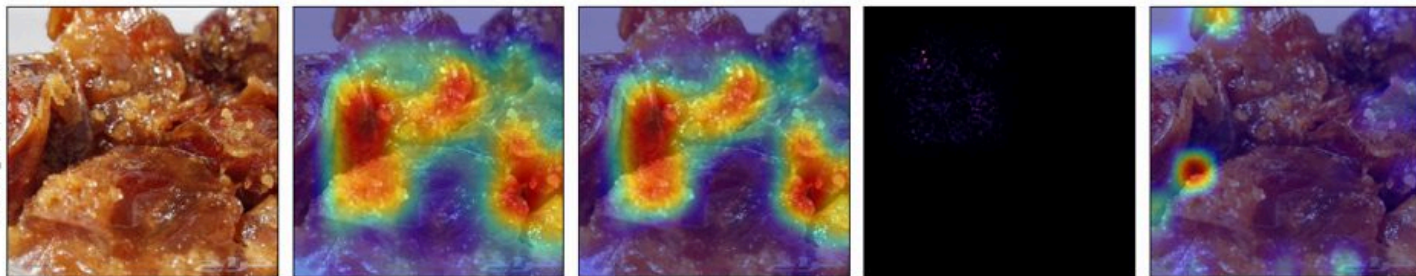
Al-shari
→ Al-shari 0.92 ✓



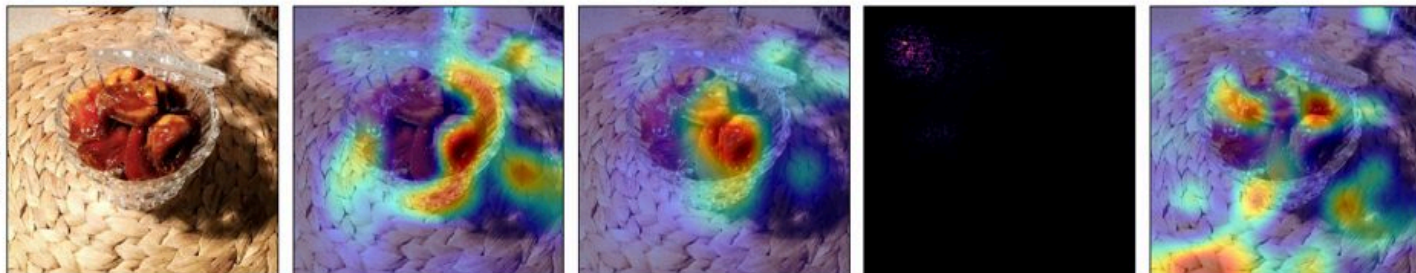
Al-sikri
→ Al-sikri 0.92 ✓



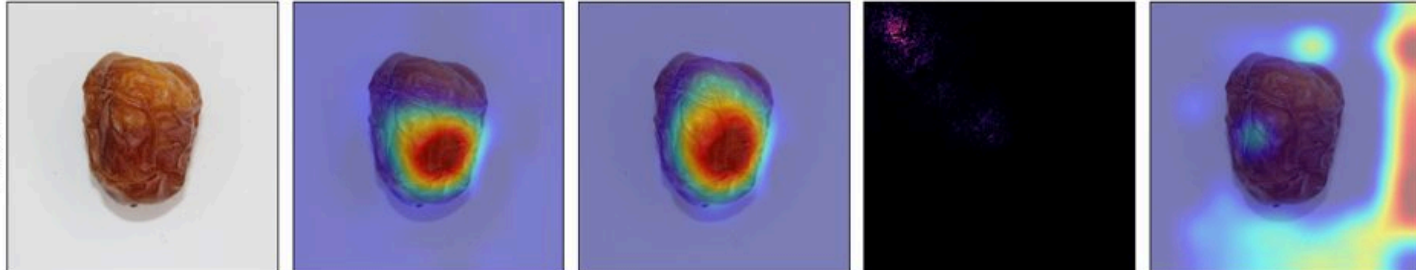
Al-sikri_magrosh
→ Al-sikri_magrosh 0.96 ✓



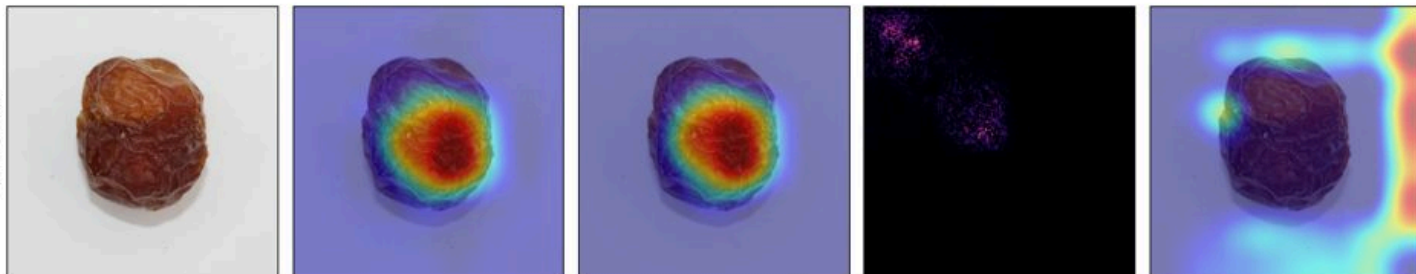
Al-sikri_magrosh
→ Al-majdool 0.26 ✗



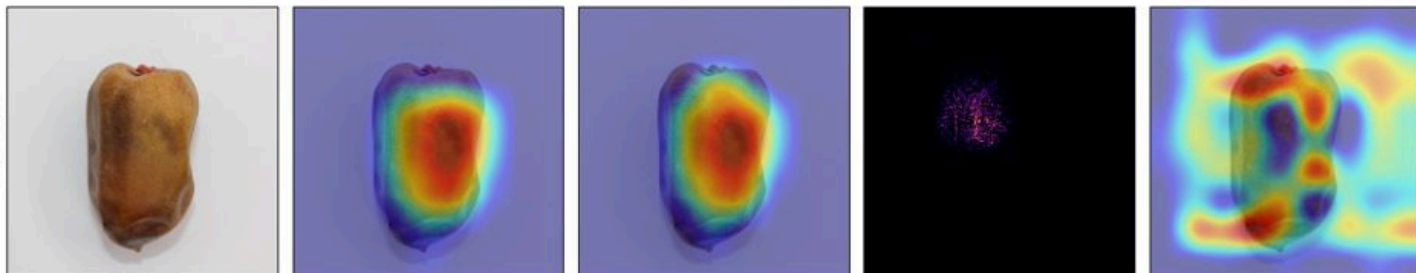
Galaxy
→ Galaxy 0.82 ✓

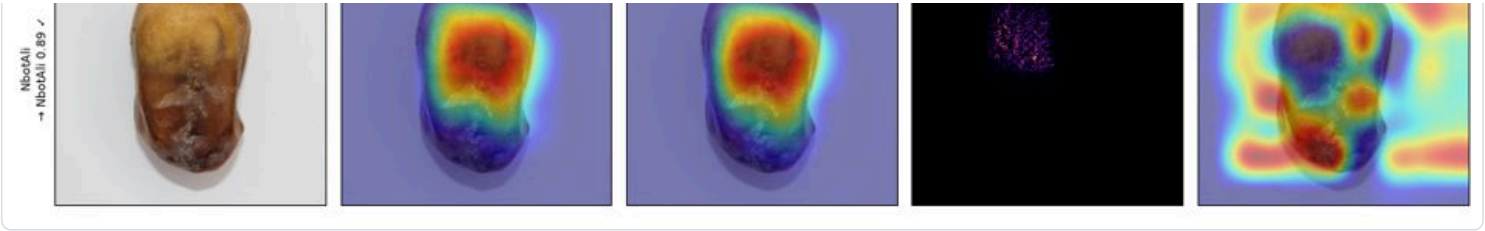


Galaxy
→ Galaxy 0.89 ✓



Nodali
→ Nodali 0.89 ✓





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
EfficientNetV2-B0 260	11.81	71.28	100.0% (n=200)	0.0033	69.2 ms

Phase 1 · Colour, shape and texture per variety

Extracted from the segmented fruit (GrabCut / plain-background mask) of every training image; the table shows each variety's median. `python src/predict.py <image>` reports the same features for one photo next to the predicted variety's typical values, with the Grad-CAM heatmap. Shape values are scale-free ratios; hue is the circular mean of skin pixels; wrinkle index is the edge density inside the fruit.

Variety	Colour	Hue °	Brightness	Saturation	Colour uniformity	Shape	Aspect ratio	Circularity	Solidity	GLCM homogeneity	GLCM contrast	LBP entropy	Wrinkle index
Al-ajwa	black / very dark brown	14	0.26	0.07	0.40	oval	1.50	0.81	0.985	0.551	5.07	3.19	0.103
Al-majdool	amber brown	15	0.52	0.51	0.37	oblong / elongated	1.75	0.76	0.987	0.552	5.24	3.17	0.112
Al-meneifi	amber brown	17	0.54	0.58	0.34	oblong / elongated	1.65	0.79	0.989	0.662	2.09	3.15	0.038
Al-muraaya	golden brown	34	0.54	0.65	0.21	oval	1.40	0.57	0.898	0.721	2.22	3.06	0.032
Al-sagai	amber brown	22	0.57	0.64	0.33	oblong / elongated	1.78	0.75	0.986	0.648	2.27	3.16	0.040
Al-shaishe	amber brown	24	0.51	0.46	0.35	oblong / elongated	1.60	0.80	0.988	0.658	2.08	3.17	0.036
Al-skari	amber brown	25	0.56	0.59	0.31	oval	1.44	0.80	0.986	0.657	1.97	3.19	0.029
Al-skri_magrosh	amber brown	24	0.62	0.64	0.23	oval	1.28	0.60	0.924	0.580	5.48	3.04	0.106
Galaxy	amber brown	15	0.50	0.56	0.33	oval	1.39	0.81	0.985	0.626	2.79	3.18	0.053
NbotAli	amber brown	23	0.56	0.62	0.34	oblong / elongated	1.64	0.76	0.984	0.703	1.46	3.15	0.014

Phase 2 · Quality grade (1–3)

Stratified split of the Kaggle grading set (4 varieties). Aseel and Fasli Toto only have Grade-1, so the informative scores are Gajar and Kupro.

84.5%

test accuracy

0.813

macro-F1

0.848

weighted-F1

0.371

test loss (cross-entropy)

98.9%

top-2 accuracy

451

test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	87.4%	0.853	0.349	84.9%
EfficientNetV2-B0 260	88.2%	0.860	0.332	84.7%
Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260 ✓ selected	89.8%	0.881	0.330	84.5%

Selected on validation: **Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260**. ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

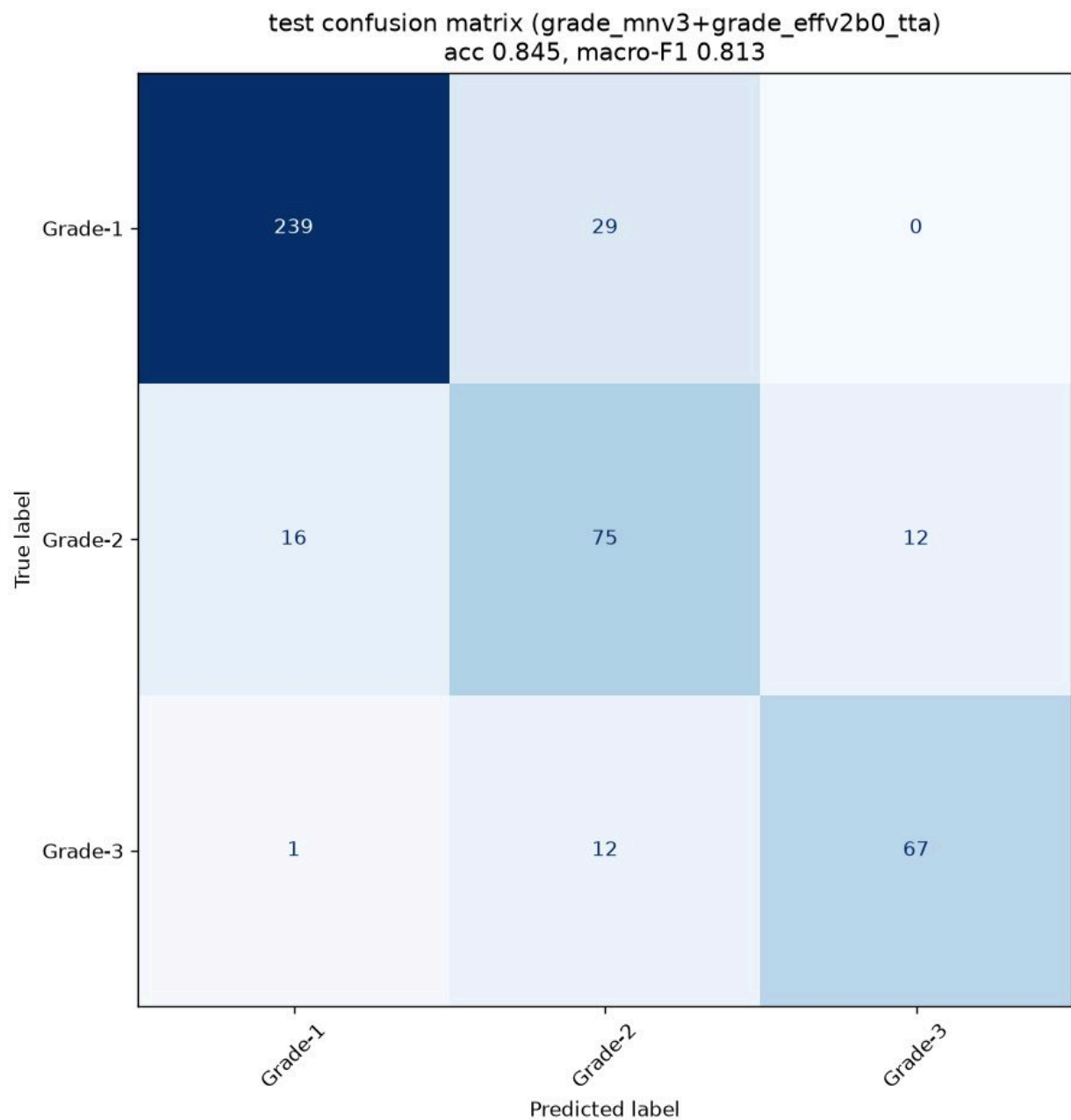
Per-class results (test)

Class	Precision	Recall	F1	n
Grade-1	0.934	0.892	0.912	268
Grade-2	0.647	0.728	0.685	103
Grade-3	0.848	0.838	0.843	80

Accuracy by source

Source / variety	Accuracy
Aseel	100.0%
Fasli-Toto	100.0%
Gajar	72.2%
Kupro	87.6%

Confusion matrix (test)



Calibration and manual-review threshold

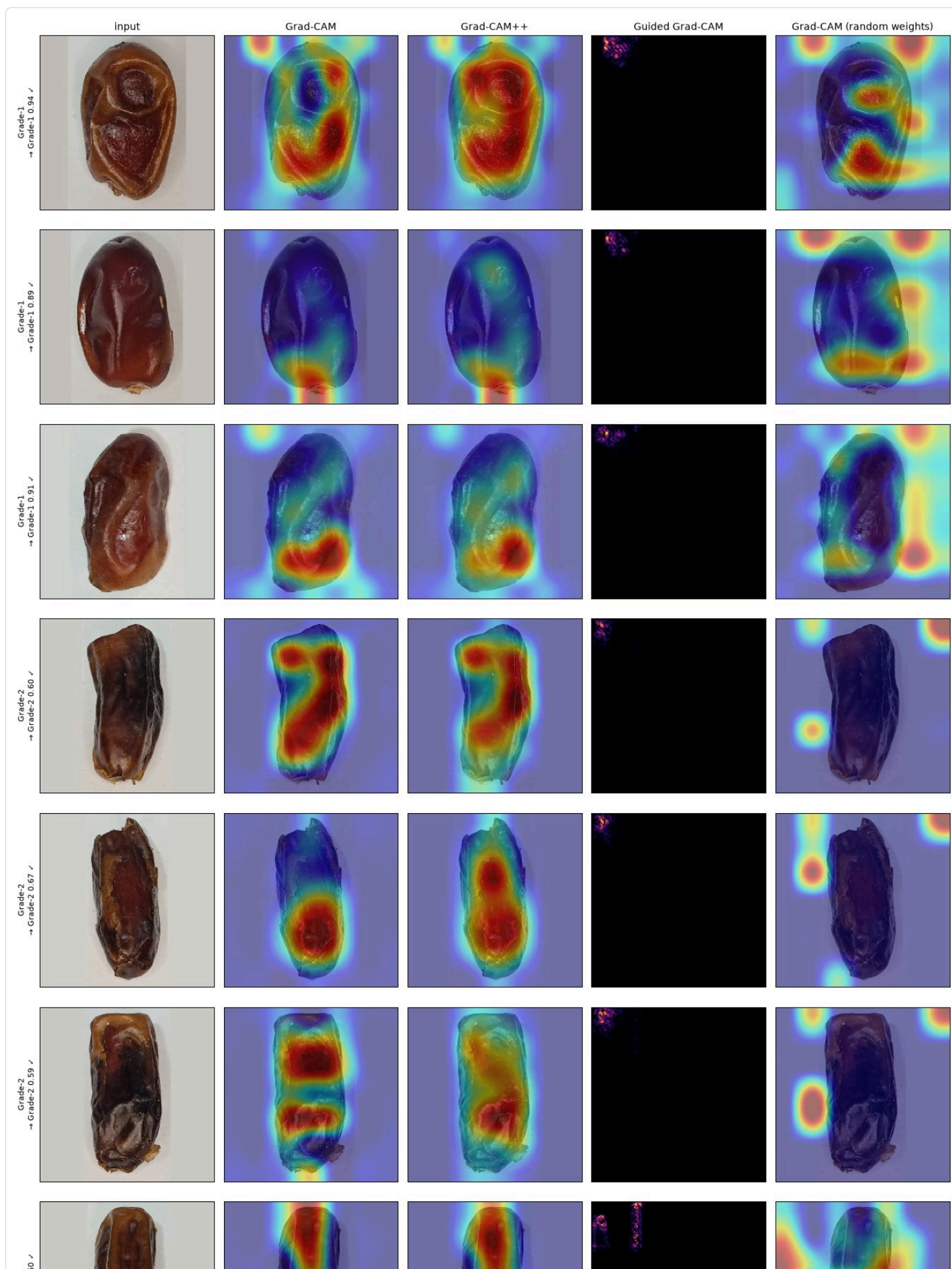
Temperature scaling ($T = 0.5486$) fitted on validation. Expected calibration error on test $0.080 \rightarrow \mathbf{0.041}$; test NLL $0.371 \rightarrow 0.343$. The review threshold (0.973) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **47.0%** of images at **97.6%** accuracy and sends 239 of 451 to manual review.

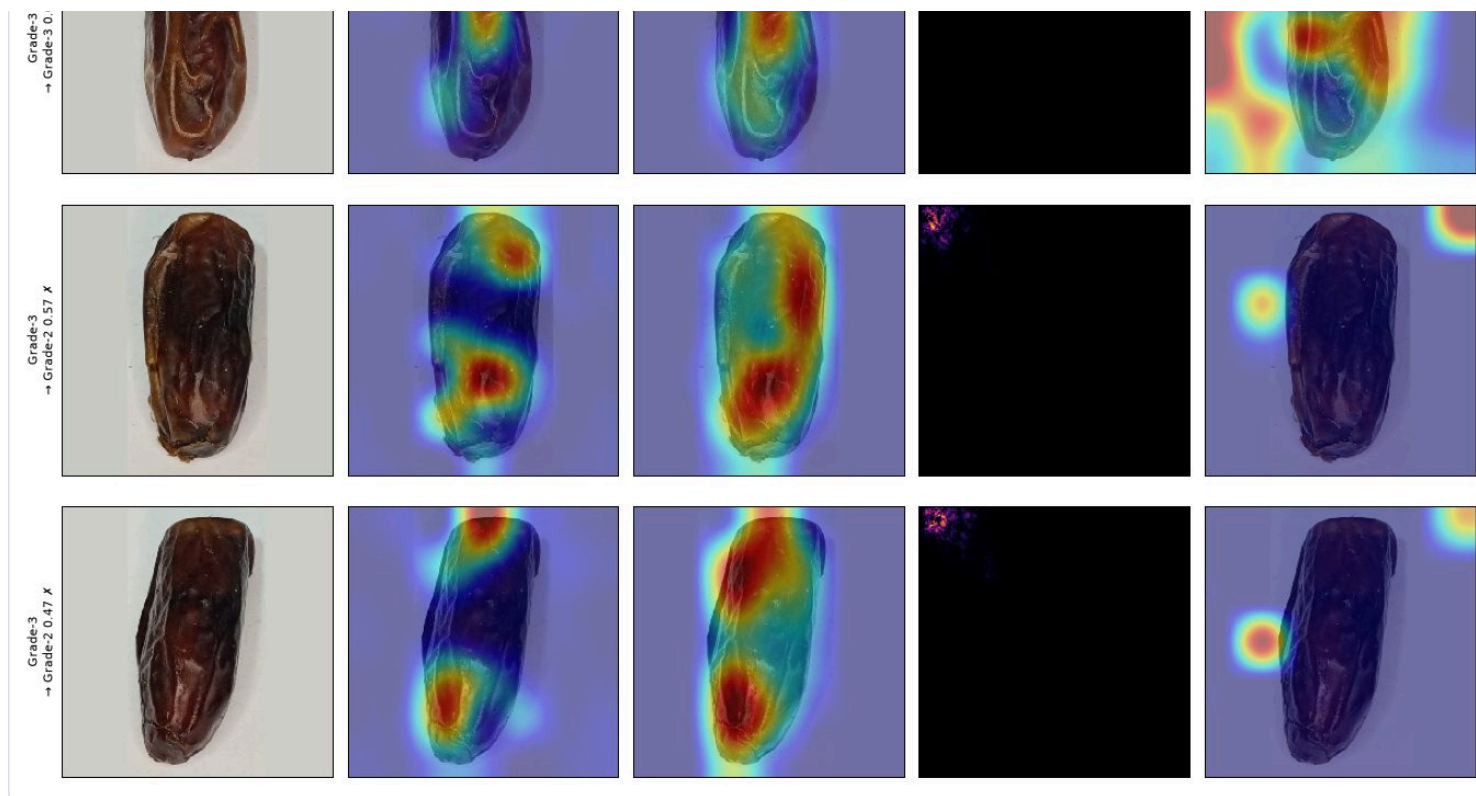
Confidence $\geq$	Auto-accepted	Accuracy on accepted
0.00	100.0%	84.5%
0.50	99.8%	84.7%
0.60	91.8%	88.6%
0.70	84.5%	91.9%
0.80	75.2%	95.6%
0.85	70.3%	95.9%
0.90	65.0%	97.3%
0.95	54.1%	98.0%
0.98	42.6%	98.4%

Grad-CAM checks below are for the ensemble member MobileNetV3-Large 224.

Explainability: Grad-CAM sanity checks (test split)

Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	-0.022	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.245 vs 0.021	pass
Share of heatmap energy on fruit pixels vs fruit share of image	82.4% vs 38.6%	pass





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
MobileNetV3-Large 224	6.0	36.36	100.0% (n=200)	0.013	13.8 ms
EfficientNetV2-B0 260	11.79	71.17	100.0% (n=200)	0.003	68.5 ms

Phase 2 · Size class (Small / Medium / Large)

80.3%

test accuracy (3 varieties with 3 sizes)

Per-variety random forest on the measured fruit outline: $\sqrt{\text{area}}$, length and width. Chosen on the validation split over logistic regression, gradient boosting and area-only cut points (val 0.839 vs 0.805–0.833). Size classes are relative to the variety (a Small Gajar is longer than a Large Aseel). Pixel models only hold for the grading-set camera rig; with an ArUco marker in the photo, size is measured in mm and classified with mm cut points.

Variety	Test accuracy	n
Aseel	98.6%	73
Fasli Toto	89.8%	59
Gajar	70.7%	198
Kupro	only Large in dataset	–

Aseel

	pred Small	pred Medium	pred Large
true Small	25	0	0
true Medium	1	24	0
true Large	0	0	23

Fasli Toto

	pred Small	pred Medium	pred Large
true Small	11	2	0
true Medium	2	19	2
true Large	0	0	23

Gajar

	pred Small	pred Medium	pred Large
true Small	67	12	4
true Medium	32	14	5
true Large	0	5	59

Gajar Small and Medium overlap almost completely in measured size (median $\sqrt{\text{area}}$ 347 vs 355 px), so the labels themselves cap accuracy for Gajar; more model capacity does not help (all four methods: 0.66–0.71).

Phase 2 · Maturity stage

Test = the whole Moroccan variety Kholt, never seen in training (each stage was shot in one session, so a random split would leak session lighting), plus held-out Alhamdan single fruits.

67.8%
test accuracy

0.667
macro-F1

0.665
weighted-F1

0.881
test loss (cross-entropy)

86.4%
top-2 accuracy

463
test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	93.2%	0.931	0.279	70.0%
EfficientNetV2-B0 260 ✓ selected	95.6%	0.956	0.206	67.8%
Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260	95.6%	0.956	0.229	not run

Selected on validation: **EfficientNetV2-B0 260**. ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

Validation (seen varieties, ~95 %) does not predict accuracy on a new variety (~65–70 %): the validation winner scores slightly *lower* on the unseen test variety than the MobileNetV3 candidate. It was kept, because switching on the test result would make the reported number optimistic. On studio photos of single fruits (Alhamdan, Rutab vs Tamar) it is 100 % (45/45).

Per-class results (test)

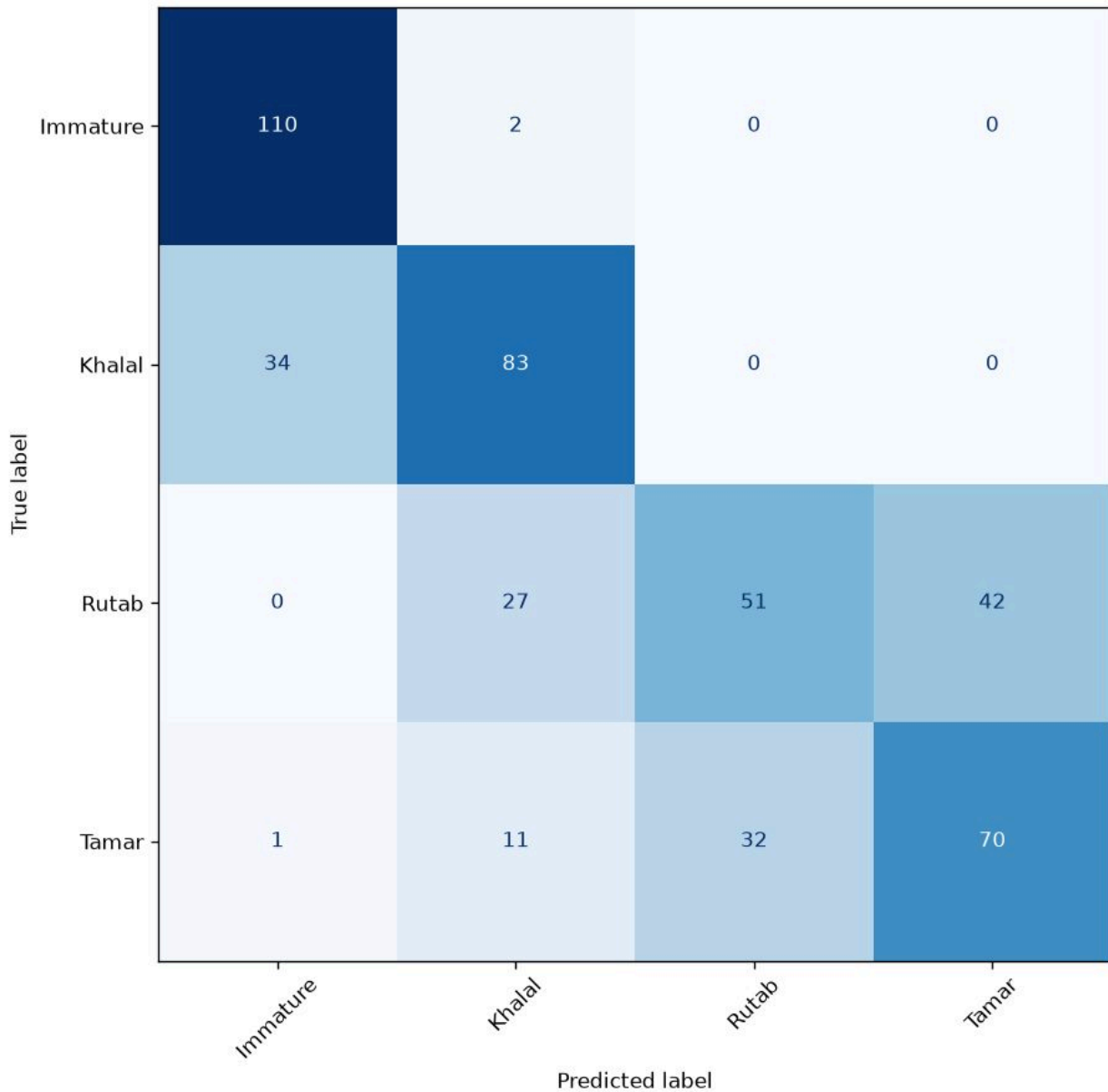
Class	Precision	Recall	F1	n
Immature	0.759	0.982	0.856	112
Khalal	0.675	0.709	0.692	117
Rutab	0.614	0.425	0.502	120
Tamar	0.625	0.614	0.619	114

Accuracy by source

Source / variety	Accuracy
alhamdan	100.0%
morocco-kholt	64.4%

Confusion matrix (test)

test confusion matrix (maturity_effv2b0_tta)
acc 0.678, macro-F1 0.667



Calibration does not transfer to the unseen test variety. Temperature scaling fitted on validation (seen varieties) makes test calibration worse, and the review threshold does not reach its target: on a new variety the model is confidently wrong, so its confidence cannot be used to route fruits to review.

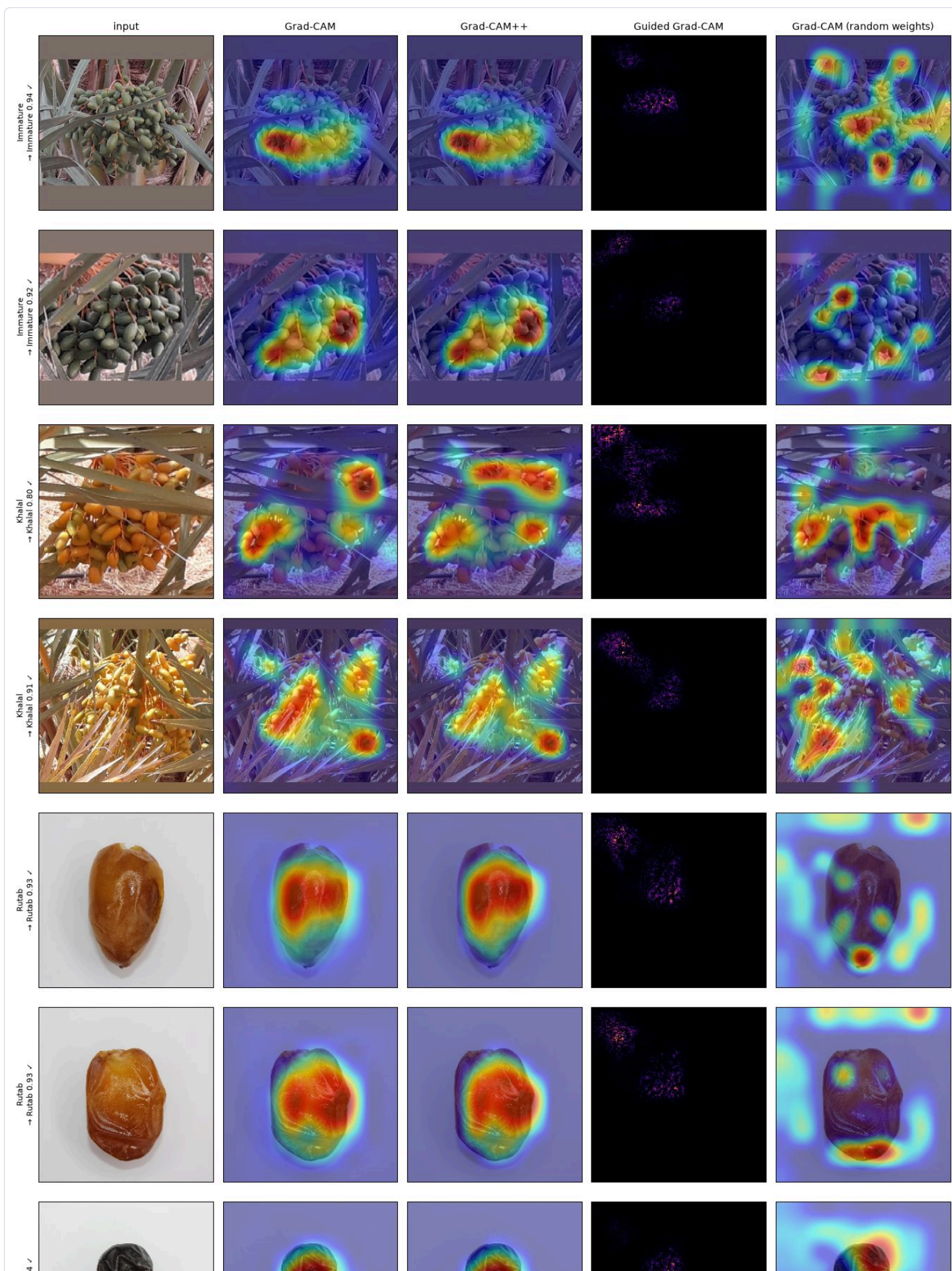
Calibration and manual-review threshold

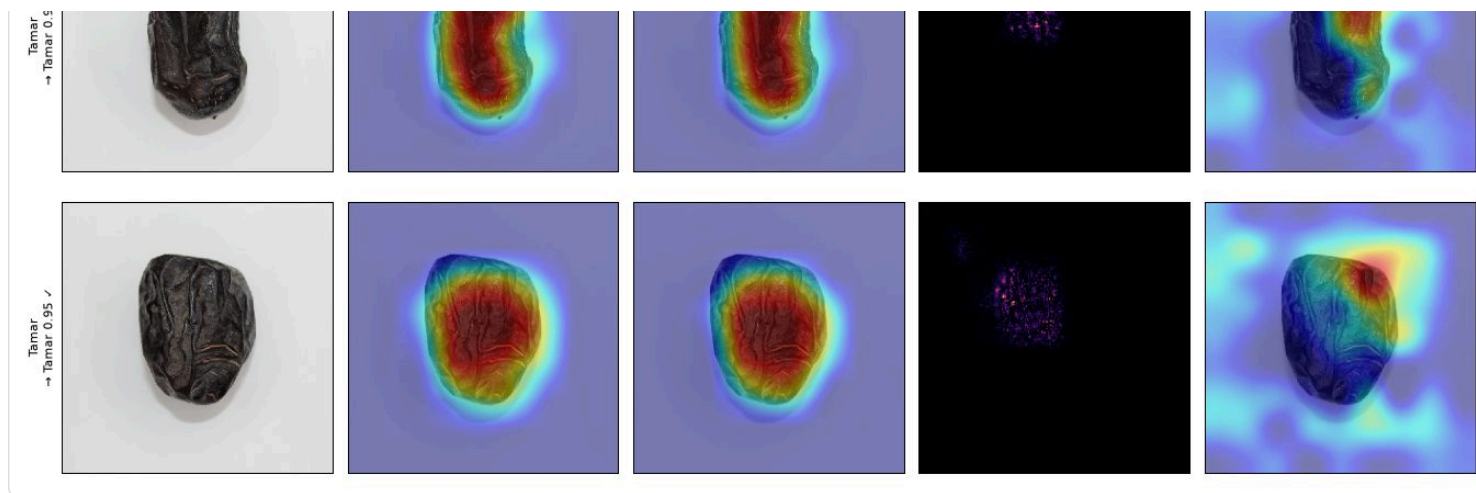
Temperature scaling ($T = 0.5358$) fitted on validation. Expected calibration error on test $0.136 \rightarrow \mathbf{0.236}$; test NLL $0.881 \rightarrow 1.291$. The review threshold (0.953) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **65.7%** of images at **81.2%** accuracy and sends 159 of 463 to manual review.

Confidence $\geq$	Auto-accepted	Accuracy on accepted
0.00	100.0%	67.8%
0.50	97.2%	68.2%
0.60	94.0%	69.0%
0.70	88.1%	71.8%
0.80	82.5%	74.6%
0.85	79.5%	75.8%
0.90	73.2%	79.1%
0.95	66.7%	80.3%
0.98	56.2%	84.6%

Explainability: Grad-CAM sanity checks (test split)

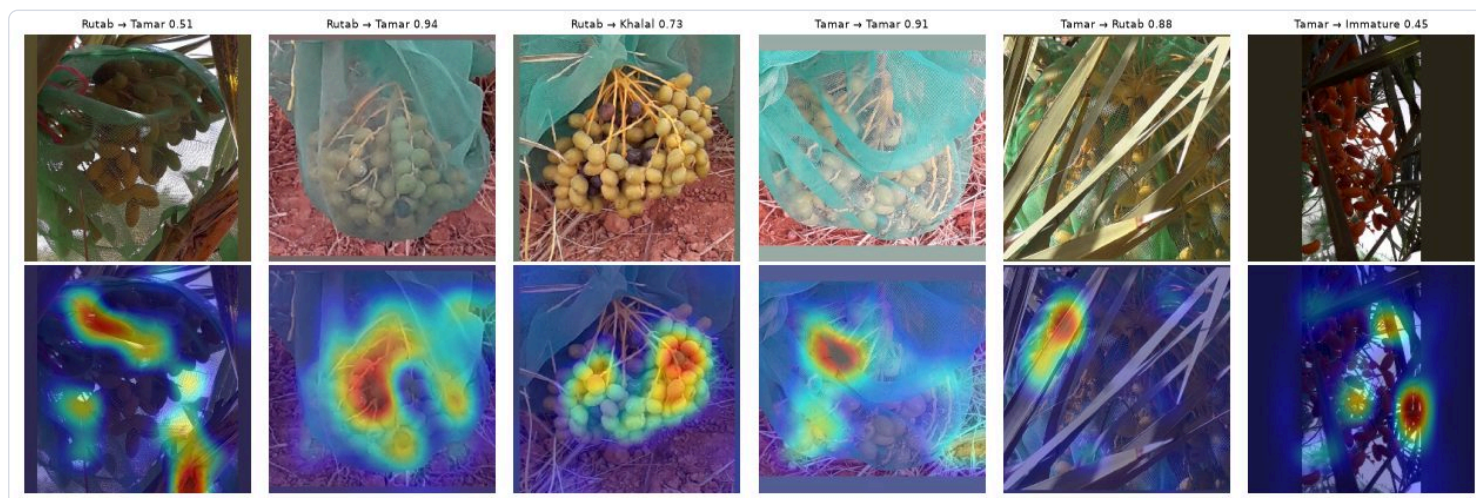
Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	0.135	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.204 vs 0.079	pass
Share of heatmap energy on fruit pixels vs fruit share of image	76.0% vs 39.9%	pass





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

Does the model look at the protective bags?



Grad-CAM on bagged Rutab/Tamar bunches of the unseen test variety: the heatmaps fall on the fruit visible through the netting, not on the net, so the bags are not being used as a shortcut. Most of these examples are misclassified, and the reason is visible: labels are per photo session, so a "Rutab" bunch still holds many green/yellow Khalal-coloured fruits, and a red "Tamar" bunch looks like another variety's earlier stage. Per-fruit stage labels on the dealer's own varieties are what would fix this.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
EfficientNetV2-B0 260	11.79	71.18	100.0% (n=200)	0.0039	67.8 ms

Phase 2 · Colour, defects, skin, weight (measurement rules)

No public dataset labels date defects, skin condition or weight, so these features are computed with transparent colour/texture rules on the segmented fruit rather than trained models. They cannot be scored with accuracy yet. As a sanity check, they are compared with the grade labels of the two varieties that have all three grades: defect area should rise from Grade-1 to Grade-3 (positive Spearman ρ with the grade number).

Feature	Gajar, Grade-1	Gajar, Grade-2	Gajar, Grade-3	Kupro, Grade-1	Kupro, Grade-2	Kupro, Grade-3	ρ Gajar	ρ Kupro
crack_pct	0.007	0.033	0.157	0.009	0.007	0.041	+0.31	+0.08
black_spot_pct	0.008	0.056	0.259	0.006	0.004	0.020	+0.48	+0.06
insect_hole_pct	0.011	0.043	0.186	0.003	0.003	0.011	+0.59	+0.15
sunburn_pct	0.000	0.000	0.000	0.014	0.000	0.000	–	-0.04
mold_pct	0.105	0.042	0.068	1.586	1.084	0.368	-0.05	-0.21
total_defect_pct	0.132	0.175	0.670	1.618	1.099	0.441	+0.53	-0.18
sugar_spot_pct	1.101	1.147	1.190	0.966	1.207	1.665	+0.09	+0.28
wrinkle_index	0.076	0.075	0.061	0.049	0.058	0.069	-0.25	+0.35
gloss_index	0.129	0.058	0.024	0.511	0.342	0.147	-0.31	-0.23
color_uniformity	0.403	0.402	0.402	0.384	0.382	0.373	-0.01	-0.08

What each feature measures

Feature	Method	Validation status
Colour uniformity	Mean CIELAB ΔE of skin pixels from their mean $\rightarrow$ score $\exp(-\Delta E/15)$; share within ΔE 10; dominant colours (k-means)	Deterministic measurement
Cracks / black spots / insect holes	Pixels ≥ 12 L* darker than their local neighbourhood; blobs split by elongation and circularity	ρ with grade +0.31 / +0.48 / +0.59 on Gajar; weak on Kupro
Sunburn / mold	Light, desaturated regions away from the rim and specular highlights; mold = compact, textured or greenish	Not validated; mold threshold raised to 2 % after false positives on glossy Kupro
Wrinkling	Edge density inside the fruit	Rises with grade on Kupro (+0.35), not Gajar (–0.25)
Gloss	Share and contrast of highlights relative to the fruit's own brightness	Falls with grade on both (–0.31, –0.23)
Sugar spots	Small light, desaturated granular blobs	Weak (+0.09 / +0.28)
Maturity by colour	Green share $\rightarrow$ Immature; yellow/red share $\rightarrow$ Khalal / partly ripened Rutab	Cannot split Rutab from Tamar (38 % on Alhamdan), so it only complements the CNN
Weight	Ellipsoid of revolution: $k \cdot L \cdot W^2$ with $k = 6.8 \times 10^{-4}$ g/mm ³ ($\rho \approx 1.3$ g/cm ³); needs an ArUco marker or mm/px	Untested on weighed fruit. A bag's scale reading recalibrates k per batch (--bag-weight)

Production readiness

Item	Status
Inference API	<code>grade.load_models()</code> + <code>grade.grade_image(path)</code> → JSON report; CLI <code>python src/grade.py img1 img2 ...</code>
Automated tests	16 passed in 30.78s (pytest tests/, 2026-09-29)
Input validation	Missing/unreadable/tiny images, images without fruit and invalid scale or bag weight raise <code>ValueError</code> ; the CLI reports each failure and exits non-zero
Confidence handling	Temperature-calibrated probabilities; predictions below the review threshold are flagged <code>needs_review</code> per fruit
Multiple fruits	Touching fruits are split at their neck (watershed); 451/451 grading test images counted as exactly one fruit
Pinned environment	<code>requirements-lock.txt</code> (Python 3.12, TensorFlow 2.18.1 + tensorflow-metal)
Edge deployment	TFLite float16 export with Keras parity check (<code>scripts/export_tflite.py</code>)
Reproducibility	Fixed seeds, split files saved (<code>data/splits.csv</code> , <code>data/quality/*_index.csv</code>), every number in this report regenerated from metric files

Limitations and what is needed before field use

1. **No dealer photos yet.** All scores are on public datasets. Eight of the requested varieties (Barhi, Mabroum, Khasab, Khasab Khalal, Helwa, Helwa Macnooz, Maktoumi, Bowyth, Kelas, Rashudia) are not in any public dataset. Accuracy on the dealer's own camera setup must be measured on a labelled sample of their photos before relying on it.
2. **Maturity data is orchard bunches.** Immature/Khalal/Rutab/Tamar labels are per bunch photo; many Rutab/Tamar bunches are inside protective bags. No public single-fruit Khalal images exist.
3. **Defects, skin and weight are unvalidated rules.** They need a few hundred dealer photos with marked defects, and 20–30 fruits weighed on a scale and photographed with the marker.
4. **Grade and size labels are subjective.** Most grade errors are between adjacent grades (1↔2); Gajar size classes overlap in measured size.
5. **99 % accuracy** is reached only on auto-accepted predictions (see the review thresholds); the remaining images go to a person. Headline accuracies above are over all test images.